

Type I and Type II Errors

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Introduction

Two kinds of error arise in hypothesis testing: rejecting a true null (Type I) and failing to reject a false null (Type II). Their probabilities – α and β – are the core quantities of statistical decision theory. Power $1 - \beta$ is one minus the Type II rate, and the rate at which sample size and effect size trade against α is the currency of study design.

Prerequisites

Hypothesis testing, p-values, sampling distributions.

Theory

The 2x2 decision table:

	H_0 true	H_0 false
Reject H_0	Type I (α)	correct (power)
Fail to reject H_0	correct	Type II (β)

- $\alpha = P(\text{reject } H_0 \mid H_0)$ is the significance level, usually set to 0.05.
- $\beta = P(\text{fail to reject } H_0 \mid H_1)$ is the Type II error rate.
- Power = $1 - \beta$, usually targeted at 0.80 or 0.90.

For a given test, power depends on: the true effect size (larger $\Rightarrow$ higher power), the sample size (larger $\Rightarrow$ higher power), the significance level (α smaller $\Rightarrow$ power lower), and the variability of the outcome (larger SD $\Rightarrow$ lower power).

Trade-off: tightening α (to 0.01, say) reduces Type I errors but increases Type II errors for a fixed sample size. Only increasing n reduces both.

Assumptions

The test is correctly specified; the distributional model holds under H_0 and H_1 .

R Implementation

```
library(ggplot2)
set.seed(2026)

# Simulation: Type I rate at alpha = 0.05 when H0 true
n_sims <- 10000
p_null <- replicate(n_sims, t.test(rnorm(30, 0, 1), mu = 0)$p.value)
mean(p_null < 0.05)

# Power at alpha = 0.05 when H1 true (effect size = 0.5)
```

```

p_alt <- replicate(n_sims, t.test(rnorm(30, 0.5, 1), mu = 0)$p.value)
mean(p_alt < 0.05)

# Compare to theoretical power
power.t.test(n = 30, delta = 0.5, sd = 1, type = "one.sample")

# Power curve
d_grid <- seq(0, 1, by = 0.05)
pw <- sapply(d_grid, function(d)
  power.t.test(n = 30, delta = d, sd = 1, type = "one.sample")$power)

ggplot(data.frame(d = d_grid, power = pw), aes(d, power)) +
  geom_line(colour = "#2A9D8F", linewidth = 1) +
  geom_hline(yintercept = 0.8, linetype = "dashed", colour = "#F4A261") +
  labs(x = "Effect size d", y = "Power",
       title = "Power curve at alpha = 0.05, n = 30") +
  theme_minimal()

```

Output & Results

```

[1] 0.0510    # empirical Type I rate
[1] 0.7549    # empirical power at d = 0.5

```

Empirical Type I error is close to the nominal 0.05; empirical power at $d = 0.5$ matches the theoretical calculation.

Interpretation

In study design, sample size is chosen to achieve a pre-specified power at the minimum effect size of interest. Reporting post-hoc power based on observed effects is misleading and discouraged.

Practical Tips

- Power at 80 % is the conventional minimum; 90 % for confirmatory trials.
- Type I and II errors are fixed by the design, not by the data; a significant p-value does not “justify” the design after the fact.
- In multiple-comparison settings, family-wise error rate replaces single-test α .
- Sequential designs (group-sequential trials) adjust α across looks to preserve overall Type I rate.
- Heavy-tailed or non-normal data reduce power; use rank-based tests or robust alternatives when applicable.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests

- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests